

Dr.-Ing. Danish IQBAL

Computational Engineer | Numerical Methods & Scientific Software

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🇪🇺 Niederlassungserlaubnis



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PROFILE

Dr.-Ing. in Computational Mechanics (magna cum laude, 1.0) with seven years in finite element methods and multi-physics simulation. Four peer-reviewed journal papers with 135+ citations and conference talks at GAMM, WCCM, GACM, and Baustatik-Baupraxis. Industry experience architecting production software with applied LLM integration. Permanent resident of Germany.

PROFESSIONAL EXPERIENCE

07/2025 - present

Software Engineer

CleverBuy GmbH, Krefeld, Germany

- Architected and shipped a dynamic-pricing engine covering 30,000+ SKUs on Refurbed and Back Market, for both sell-side listings and buyback offers.
- Reduced the operations lead's manual pricing work by 80% with a real-time monitoring and override dashboard (Django/Vue).
- Designed a priority-based Celery job engine on PostgreSQL/Django; 24/7 pricing cycles over the JTL-Wawi ERP via a service-layer schema.
- Integrated Refurbed and Back Market REST APIs, automated SKU creation in JTL with image processing, and Claude (Anthropic) API for grade/pricing decision support.

08/2017 - 03/2024

Research Assistant

University of Duisburg-Essen, Essen, Germany

- Developed and implemented polygon-based SBFEM in MATLAB (object-oriented) for transient thermoelastic fracture in functionally graded materials; verified against ANSYS.
- Authored four peer-reviewed journal papers in CMAME, Engineering Fracture Mechanics, and Theoretical and Applied Fracture Mechanics, with 135+ citations; peer reviewer for the International Journal of Non-Linear Mechanics.
- Designed and delivered graduate Structural Dynamics with ANSYS (modal, static, transient) on a university-funded inverted-classroom grant where practicing engineers analysed real vibration problems and designed tuned-mass dampers.
- Taught Computer Languages for Engineers (Fortran 77/95 OOP, C++ OOP) with numerical-methods term projects (Newton interpolation, QR decomposition).
- Supervised three master's and one bachelor's thesis spanning ANSYS CFD (sloshing, Housner approach), RFEM (footbridge dynamics), and SBFEM; TA for Abaqus with Python scripting.

PROFESSIONAL EXPERIENCE

01/2020 - 03/2020	Visiting Researcher Federation University, Ballarat, Australia Australia-Germany research cooperation (Federation University, DAAD) on polygon SBFEM for coupled thermoelastic fracture in functionally graded materials.
01/2017 - 03/2017	Student Research Assistant University of Duisburg-Essen, Essen, Germany Implemented a polygon-based SBFEM in Fortran and integrated it into the open-source FE code FEAPpv.
06/2015 - 10/2015	Seasonal Worker (Assembly Line) Mercedes-Benz Group AG, Düsseldorf, Germany Spot welding with jigs and automotive assembly-line work.
10/2014 - 12/2014	Seasonal Worker (Assembly Line) Mercedes-Benz Group AG, Rastatt, Germany Production tasks on automotive assembly line.
12/2013 - 12/2013	Seasonal Worker (Logistics) Amazon Deutschland Services GmbH, Rheinberg, Germany Warehouse scanning and packaging preparation.

CORE COMPETENCIES

- Computational Mechanics
 - Multi-Physics Simulation
- Numerical Methods
 - Scientific Software Development
- Finite Element Analysis
 - Applied LLM Integration

SKILLS

Numerical Methods:	• FEM	• SBFEM	• XFEM	• CFD	
Simulation Tools:	• ANSYS	• Abaqus	• RFEM	• Gmsh	• ParaView
Programming:	• MATLAB	• Python	• Fortran	• C++	
Backend & Data:	• Django	• PostgreSQL	• Celery	• REST APIs	
Applied AI:	• Claude API	• Gemini	• Groq	• Prompt engineering	
Infrastructure:	• Linux	• nginx	• Hetzner	• GitHub Actions	

EDUCATION

08/2017 - 07/2024	Ph.D. (Dr.-Ing.) Computational Mechanics University of Duisburg-Essen, Germany - Thesis: Development of scaled boundary polygon elements for coupled thermoelastic fracture modeling. - Grade: Magna cum laude (1.0)
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EDUCATION

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| 10/2013 - 07/2017 | <p>M.Sc. Computational Mechanics
University of Duisburg-Essen, Germany</p> <ul style="list-style-type: none">• Thesis: Development of polygon elements based on the scaled boundary finite element method.• Grade: Good (2.0) |
| 01/2009 - 12/2012 | <p>B.E. Automotive Engineering
NED University of Engineering & Technology, Karachi, Pakistan</p> <ul style="list-style-type: none">• Thesis: Design and analysis of motorcycle oil pump.• Project: Formula Student car (design, development, manufacturing).• Grade: First Division |

PERSONAL PROJECTS

sahab.pk — AI-Powered Ledger SaaS

Solo founder & architect — live alpha

- Multi-tenant SaaS for retailers in Pakistan: shopkeepers chat in Urdu or English; an LLM extracts structured events into a double-entry ledger with row-level tenant isolation.
- Django 5, PostgreSQL 18, Redis, SvelteKit, Flutter; Gemini + Groq LLMs; Hetzner + nginx + GitHub Actions CI/CD.

MATLAB Free-Body Diagram Toolbox

Open source — github.com/M-D-Iqbal

- MATLAB toolbox used in own Ph.D. thesis and in graduate Structural Dynamics course materials at the University of Duisburg-Essen.

PUBLICATIONS & CONFERENCE PRESENTATIONS

Peer-Reviewed Journal Papers

M. D. Iqbal, et al. "Transient thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." *Theoretical and Applied Fracture Mechanics*, vol. 127, Oct. 2023, p. 104056.

M. D. Iqbal, et al. "Thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." *Engineering Fracture Mechanics*, vol. 264, Apr. 2022, p. 108305.

M. D. Iqbal, et al. "Development of the scaled boundary finite element method for crack propagation modeling of elastic solids subjected to coupled thermo-mechanical loads." *Computer Methods in Applied Mechanics and Engineering*, vol. 387, Dec. 2021, p. 114106.

E.T. Ooi, M. D. Iqbal, et al. "A polygon scaled boundary finite element formulation for transient coupled thermoelastic fracture problems." *Engineering Fracture Mechanics*, vol. 240, Dec. 2020, p. 107300.

Conference Papers & Talks

C. Birk, R. Assaf, M. D. Iqbal, H. Gravenkamp. "Numerische Strukturmodellierung mit Polygon-Elementen." **Baustatik-Baupraxis 15**, 2024.

M. D. Iqbal, et al. "Scaled boundary polygon element formulation for coupled thermoelastic fracture in FGMs." **GACM Colloquium**, 2022.

M. D. Iqbal, et al. "Polygon SBFEM for transient coupled thermoelastic fracture." **WCCM XIV**, 2020 (virtual).

M. D. Iqbal, et al. "Scaled boundary finite element modelling of thermally induced crack propagation." **GAMM Annual Meeting**, 2019.

M. D. Iqbal, C. Birk. "Numerical modelling of thermally induced crack propagation in functionally graded materials using the scaled boundary finite element method." *Forschungskolloquium Baustatik-Baupraxis 2018*, Grasellenbach.

Peer Review

Reviewer, **International Journal of Non-Linear Mechanics**: "Thermally nonlinear analysis of propagating cracks under generalized thermal shock".

Theses

M. D. Iqbal. "Development of scaled boundary polygon elements for coupled thermoelastic fracture modeling." Ph.D. Thesis, University of Duisburg-Essen, 2024.

M. D. Iqbal. "Development of polygon elements based on the scaled boundary finite element method." M.Sc. Thesis, University of Duisburg-Essen, 2017.

LANGUAGES

Urdu (native)

English (C2)

German (fluent, B2)

Essen, 22 May 2026

